

Formulas

$$y = \int f(x)dx = F(x) + C$$

$$\int F'(x) dx = F(x) + C.$$

$$\frac{d}{dx} \left[\int f(x) dx \right] = f(x).$$

- $\int 0 dx = C$
- $\int dx = x + C$
- $\int k dx = kx + C$
- $\int kf(x) dx = k \int f(x)dx$
- $\int x^n dx = \frac{x^{n+1}}{n+1} + C$
- $\int [f(x) \pm g(x)] dx = \int f(x)dx + \int g(x)dx$

- $\int a^x dx = \frac{a^x}{\ln a} + C$

- $\int \frac{1}{x} dx = \ln x + C$

- $\int e^x dx = e^x + C$

$$\int_a^b f(x)dx = [F(x)]_a^b = F(b) - F(a)$$